DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING,
162 WATTS, 120 VAC, 60 HERTZ, DRIPPROOF, SYMBOLS 76.1, 76.3, 76.4, AND 76.5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

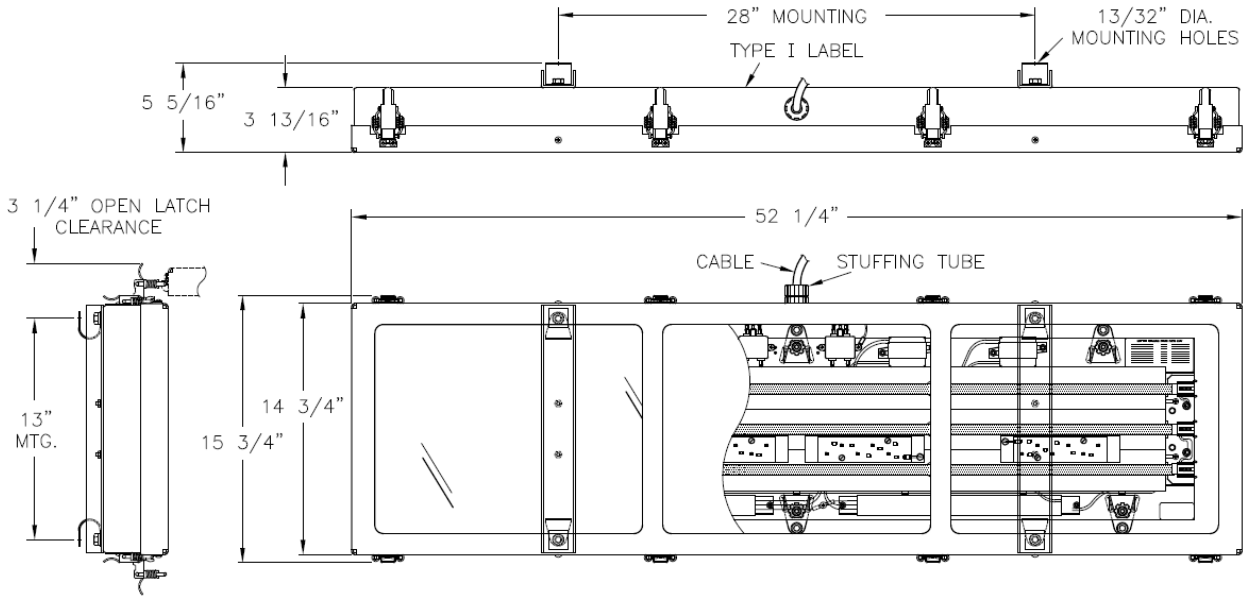
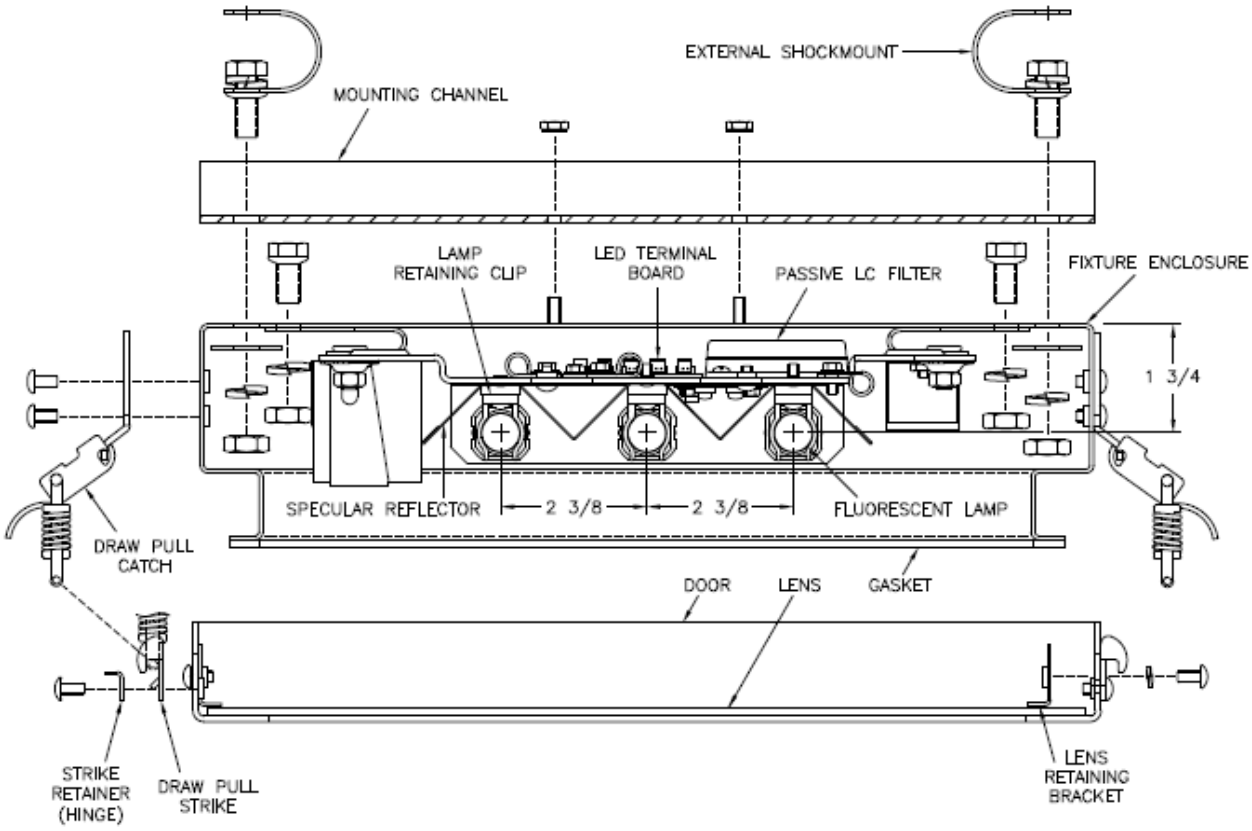


FIGURE 1. Typical views of symbol 76.4.

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NOTES:

1. Material for window shall be polycarbonate as follows:
   Part Number M16377/85-001. Clear, polycarbonate, nominal ⅛ inch thick, Type CLEAR-1 of MIL-DTL-24191.
2. Material for window gasket shall be silicone rubber (35-45 durometer). Gasket shall adhere to light fixture housing. Gasket may be furnished in segments.
3. Window assembly shall be mechanically secured to door frame assembly.

FIGURE 2. Typical section for symbol 76.4.

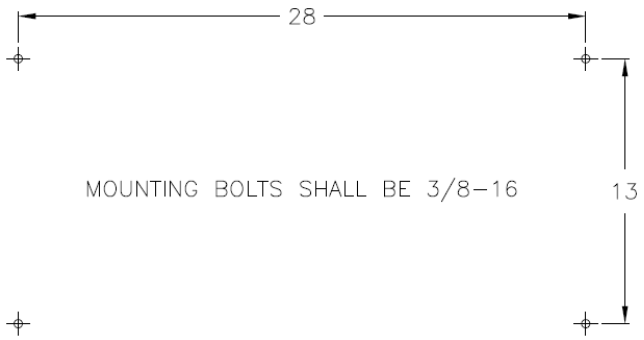


FIGURE 3. Mounting dimensions.

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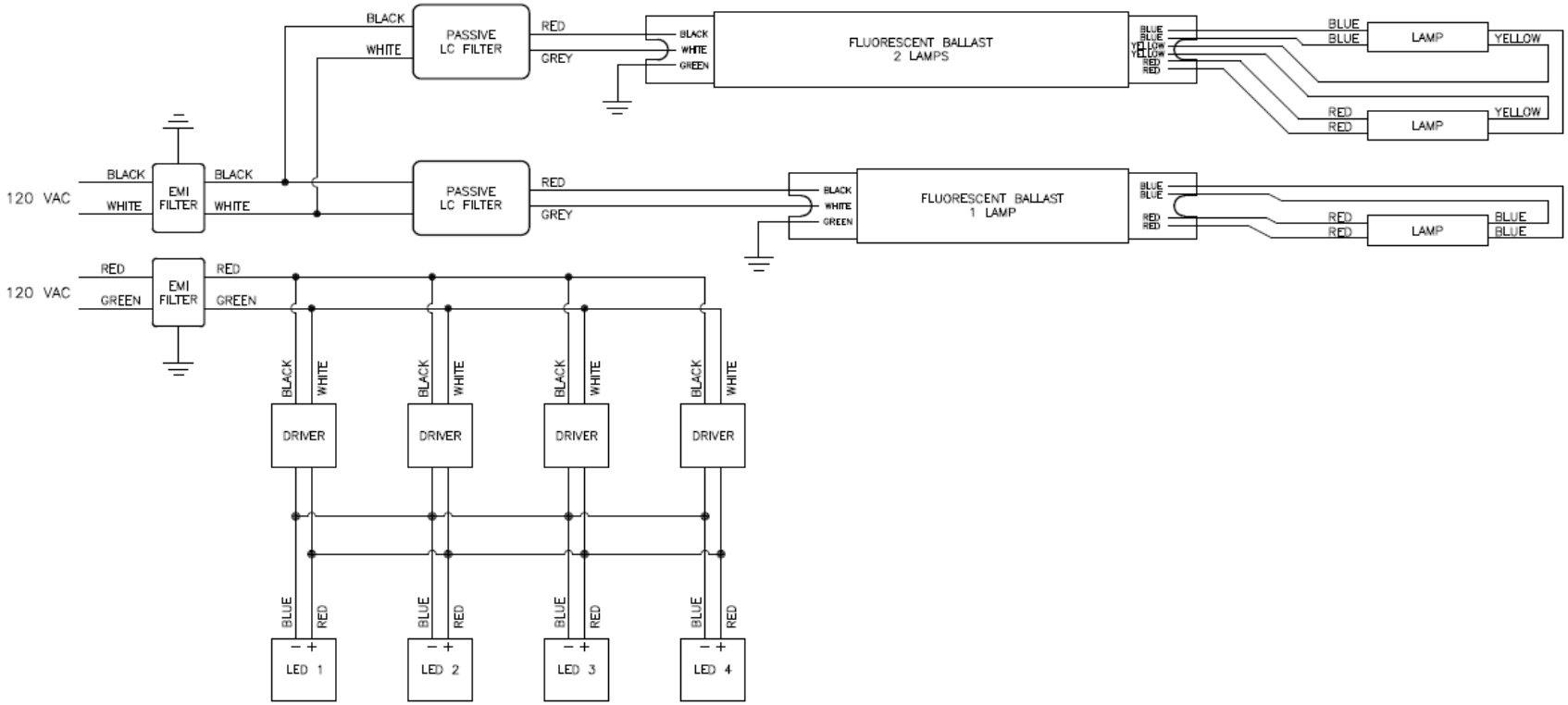


FIGURE 4. Wiring diagram for symbols 76.4 and 76.5.

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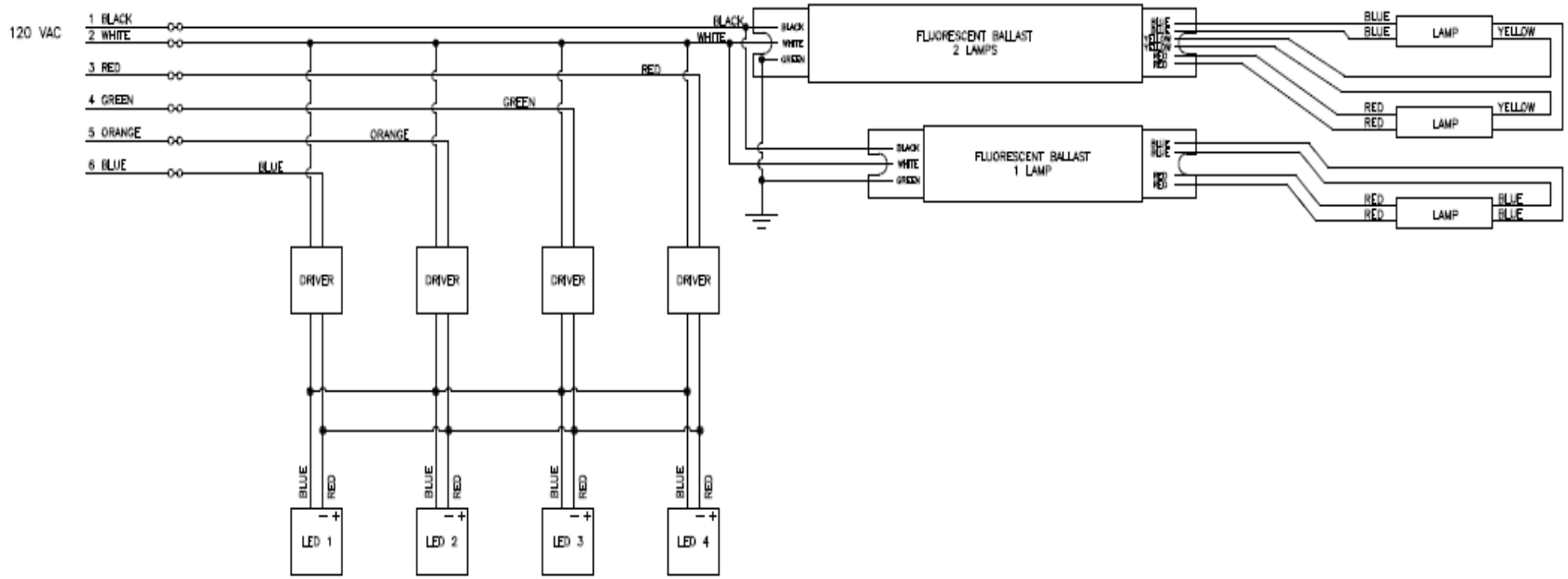


FIGURE 5. Wiring diagram for symbol 76.1.

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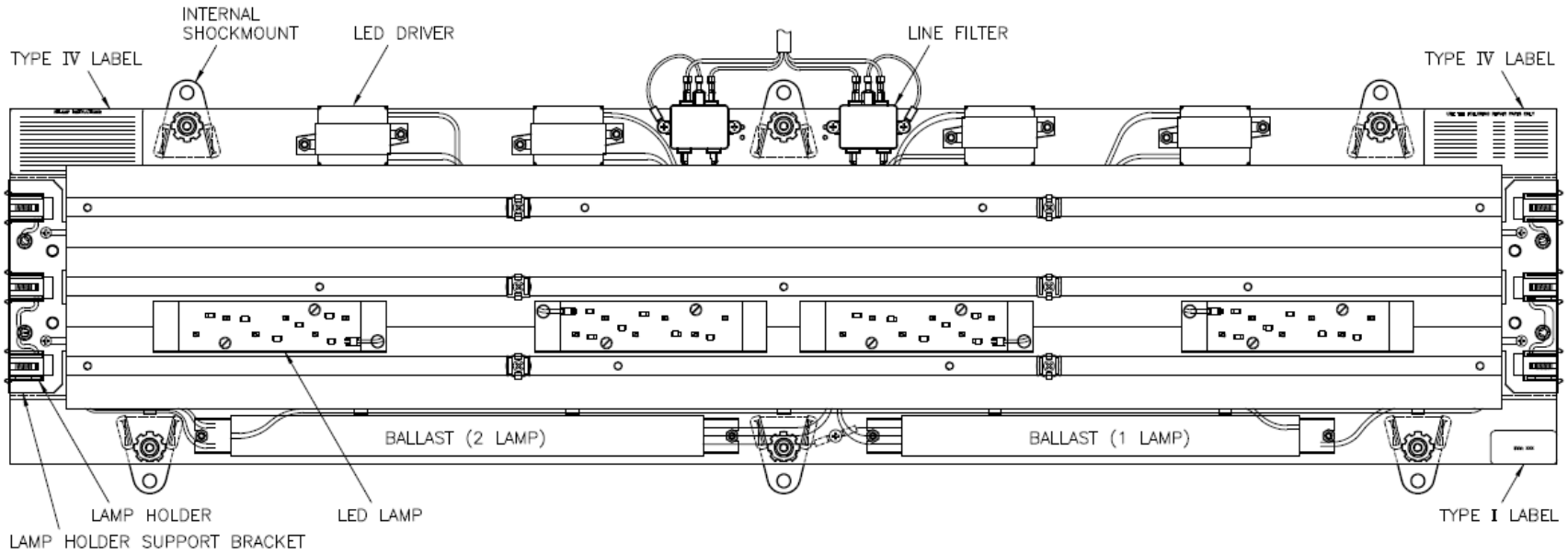
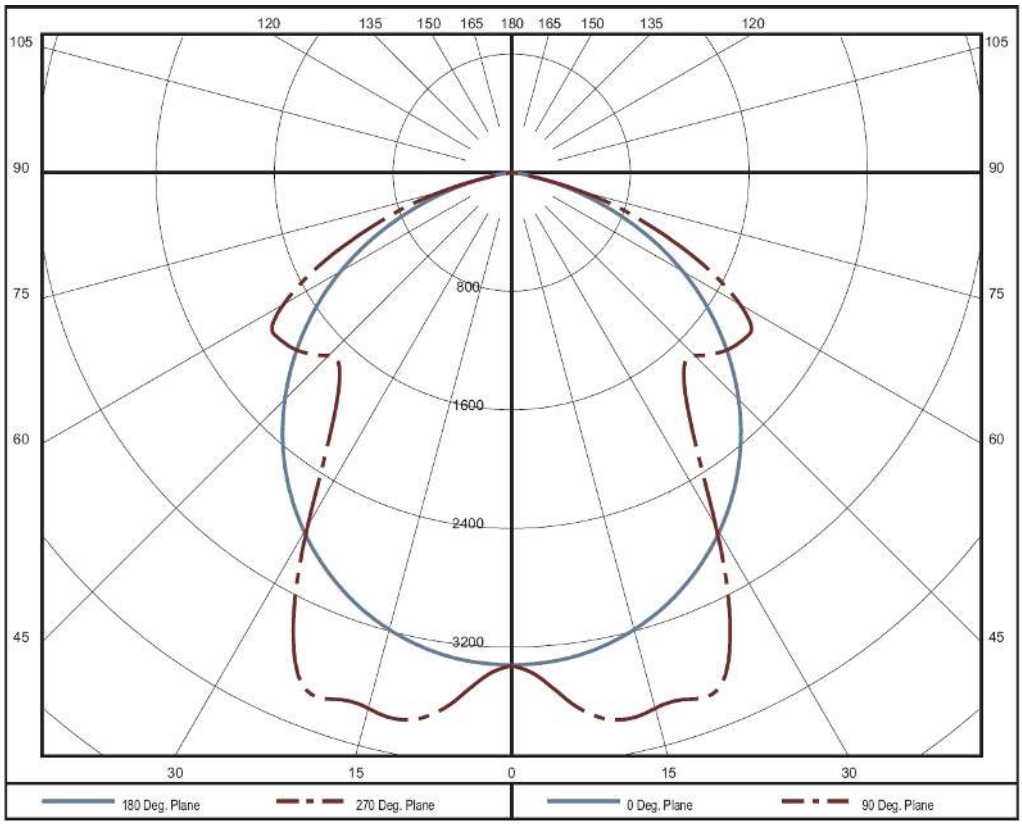


FIGURE 6. Reflector assembly for symbols 76.1, 76.4, and 76.5.

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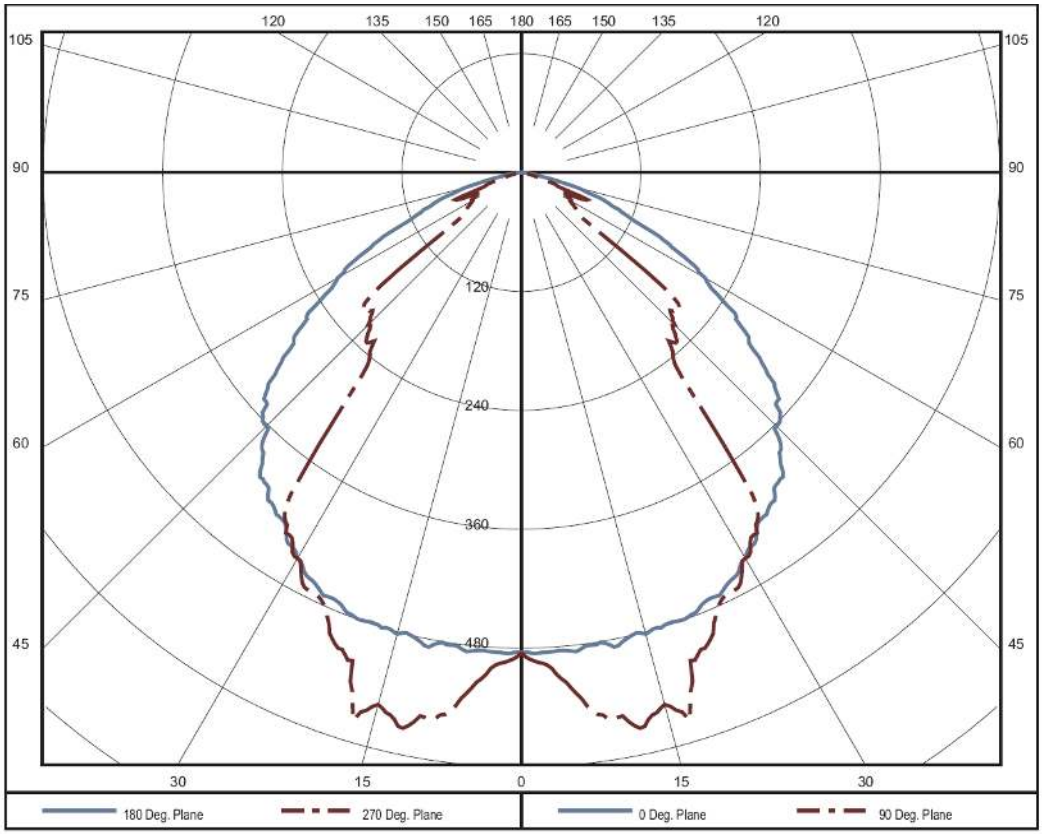


PHOTOMETRIC TEST DATA:

Distance:	32.8 feet.
Plane:	22.5-degree lateral increments and 0.5-degree vertical increments.
Voltage:	120 VAC.
Ambient temperature:	77 °F (25 °C).
Lamps:	Sylvania FP54/841C/HO/ECO.
Ballasts:	Furnished with fixture.

FIGURE 7. Candela distribution curves, white mode, symbols 76.1, 76.3, 76.4, and 76.5.

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PHOTOMETRIC TEST DATA:

Distance:	32.8 feet.
Plane:	22.5-degree lateral increments and 0.5-degree vertical increments.
Voltage:	120 VAC.
Ambient temperature:	77 °F (25 °C).
LEDs:	SSL 3.9.2.5 / 2W 700MA.
Driver:	M16377/85-003.

FIGURE 8. Candlepower distribution curves, night vision imaging system (NVIS) mode, symbols 76.1 and 76.4.

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TABLE I. Numerical candela distribution, white mode, for symbols 76.1, 76.3, and 76.4.

Vertical Angle	Candela Distribution	
	Vertical Angle	
	0-Degree	90-Degree
0	3326	3326
5	3307	3512
15	3191	3741
25	2957	3485
35	2613	2125
45	2163	1745
55	1628	1933
65	1016	1371
75	384	479
85	17	7

TABLE II. Numerical candela distribution, NVIS mode, for symbols 76.1 and 76.3.

Vertical Angle	Candela Distribution	
	Vertical Angle	
	0-Degree	90-Degree
0	485	485
5	484	516
15	482	556
25	471	476
35	424	413
45	362	218
55	267	68
65	153	51
75	55	15
85	2	0

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REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBER:

76.1: Complete lighting fixture assembly, yoke mounted, stainless steel, clear window; for use with three 54-watt 4,100K white lamps and Night Vision Friendly LEDs.

76.3: Complete lighting fixture assembly, standard mounting, aluminum, clear window; for use with three 54-watt 4,100K white lamps.

76.4: Complete lighting fixture assembly, standard mounting, aluminum, clear window; for use with three 54-watt 4,100K white lamps and Night Vision Friendly LEDs.

76.5: Complete lighting fixture assembly, standard mounting, aluminum, clear window; for use with three 54-watt 4,100K white lamps and red LEDs.

DESIGN:

Electrical rating:	162 watts maximum, 120 VAC, 60 hertz.
Dimension and configurations:	See figures 1 and 2.
Mounting dimensions:	See figure 3.
Enclosure:	
Symbol 76.1:	Stainless steel, dripproof.
Symbols 76.3, 76.4, and 76.5:	Aluminum, dripproof.
Cable:	
Symbol 76.1:	No cable provided.
Symbol 76.3:	20 feet of LSDHOF-3 in accordance with MIL-DTL-24643/3.
Symbols 76.4 and 76.5:	20 feet of LSFHOF-3 in accordance with MIL-DTL-24643/3.
Wiring diagram:	
Symbol 76.1:	See figure 5.
Symbols 76.4 and 76.5:	See figure 4.
Internal wiring:	Wire shall be of stranded soft annealed copper of a cross section to provide ample and safe current-carrying capacity and mechanical strength.
Lampholder spacing:	The distance between movable lampholder support bracket faces in closed operating position shall be 46.65±0.060 inches.
Lampholder support brackets:	Lampholder support brackets shall be provided. The bracket shall support spring loaded lampholders at each end. Each end shall be adjustable to allow for lamp replacement. Captive fasteners, or screws that do not require removal for support bracket adjustment, shall be used to permit lamp replacement. In no case shall the lamps fall from the fixture when movable support brackets are in loosened position. Secondary lamp support shall be provided by two non-metallic lamp retaining clips per lamp.
Wiring assembly:	See figure 6. Material shall be aluminum sheet, 0.090 inch thick, with a specular minimum 92 percent reflective material. Lamps, LED arrays, lampholders, wiring, LED drivers, and fluorescent ballast shall be mounted to the wiring support channel. All LED driver outputs shall be interconnected such that if one driver fails, all LED arrays remain functional.
Light output:	
Symbols 76.1, 76.3, 76.4, and 76.5:	Fluorescent, see figure 7.
Symbol 76.1 and 76.4:	Night Vision, see figure 8.

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Weight:

Symbol 76.1:	61 pounds maximum.
Symbol 76.3:	31 pounds maximum.
Symbols 76.4 and 76.5:	33 pounds maximum.

Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 1, 2, and 4:

a. Two Type I labels.

b. One Type IV label. This label shall provide clear and concise step-by-step procedures for the removal and replacement of lamps in the fixture.

c. One Type IV label. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table III and window assemblies of table VI by name and stock number only.

Standard parts: The standard parts shown in table III shall be used in the construction of the fixtures specified herein.

EXAMINATION AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article inspection:

Complete fixtures:	See table IV.
Window assemblies:	See tableV.

Conformance:

Comparison inspection:	Same as first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table VI.

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TABLE III. Standard parts.

Name	Military P/N or Specification	National Stock Number (NSN)	Remarks
Lamp	Commercial identification F54T5HO/CW		4,100K, 25,000 hours min, 4,600 mean lumens min. @ 95 °F (35 °C).
Ballast – 2 lamps	UL 935 Class P, Type CC, Type HL, Type 1 outdoor		High power factor program start
Ballast – 1 lamp	UL 935 Class P, Type CC, Type HL, Type 1 outdoor		High power factor program start
LED array, cyan			6 up flood
LED array, red			6 up flood
LED driver			24 VDC, 700ma
Shock mount, internal	M16377/41-008	6210-01-393-4302	
Shock mount, external	M16377/41-021	6210-00-396-1710	0.067 thick, 316 stainless steel
Lampholder assembly, left	UL 496		Three, spring loaded lampholders attached to bracket
Lampholder assembly, right	UL 496		Three, spring loaded lampholders attached to bracket
Line EMI filter			On incoming power
Passive LC filter			Power to fluorescent ballast

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TABLE IV. First article tests and comparison inspection for complete luminaire. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Electromagnetic interface	3.6.14	4.8.16	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/ Dripproof
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/ Dripproof
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	2/
Light output (photometric data)	3.7.1	4.8.6.1	Candlepower
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ The fixture shall be mounted in normal operating position and energized for a period of 2 hours prior to, and throughout, the water test which shall be conducted as follows: A continuous spray (similar to that of a sprinkling can) at a distance of 1 foot shall be directed to successively cover all surfaces, while fixture is slowly rotated on the long axis alternately exposing all parts of the fixture (side, bottom, side, window, and so forth). Minimum duration of test shall be 5 minutes.

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TABLE V. First article tests and comparison inspection for window assemblies. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof
NOTE: 1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.			

TABLE VI. Part numbers.

Military Part Number M16377/85-	Item Name	National Stock Number (NSN)
76.1	Complete fixture, stainless steel, white/night vision	
76.3	Complete fixture, aluminum, white	
76.4	Complete fixture, aluminum, white/night vision	
76.5	Complete fixture, aluminum, white/red	
001	Window, CLEAR-1 acrylic	
002	Window, clear polycarbonate	
003	Driver	

Custodians:
Army – CR4
Navy – SH
Air Force – 99

Review activities:
Navy – AS, CG
Air Force – 03
DLA – GS2

Civil agency:
GSA – FAS

Preparing activity:
Navy – SH
(Project 6210-2012-010)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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